Genomic & Pharmacogenomic Clinical Summary

3-PAGE PORTRAIT BRIEF (OLD)

Patient / Sample ID: Daniel Ehrle (Daniel_Ehrle) | **Assay:** 48.8x BAM mean depth Phased WGS (GRCh38.p14) | **Date:** October 12, 2026

MONOGENIC CARRIERS
3 (Het)
CBLIF, GJB2, VDR (Recessive/Dom)

PHARMACOGENOMICS
10 Priority
CYP2C19, CYP2C9, CYP2W1, CYP3A5

DISEASE MODIFIERS
4 Modifiers
Organ / Metabolic

1. PRIORITY GENOMIC FINDINGS & CARRIER STATUS

Primary Care & Specialist Triage

GENE	VARIANT & LOCUS	ZYGOSITY	CLASSIFICATION	AI IMPACT (PHRED)	CLINICAL IMPACT & PRIMARY CARE GUIDANCE
CBLIF	c.79+1G>A · rs147785187	Heterozygous	Pathogenic	AVI 33.9 / CADD 32.0 / SpliceAI 0.99	Canonical splice-donor variant (c.79+1G>A) in gastric intrinsic factor (autosomal recessive **Hereditary Intrinsic Factor Deficiency, OMIM #261000**), which mediates ileal cubam-receptor endocytosis of dietary Vitamin B12. **Clinical Impact:** Heterozygous carriers preserve intrinsic factor secretion from the wild-type allele; periodic serum Vitamin B12 (> 350 pg/mL) and methylmalonic acid (MMA) verify cellular cobalamin stores (if subclinical depletion occurs, oral/sublingual B12 1,000 mcg/day restores levels via passive mucosal diffusion).
GJB2	p.Met34Thr · c.101T>C · rs35887622	Heterozygous	Pathogenic	AVI 23.6 / CADD 20.9 / REV 0.702	Coding variant (p.Met34Thr) in cochlear gap junction protein Connexin-26 (DFNB1A, OMIM #220290), which recycles endolymphatic potassium ions in hair cells. **Clinical Impact:** Heterozygous carriers typically maintain normal hearing thresholds; establish a baseline pure-tone audiogram and exercise caution with ototoxic aminoglycoside antibiotics (gentamicin, tobramycin, amikacin) when alternative antimicrobials exist.
VDR	c.-1172A>G · rs4516035	Homozygous	Pathogenic	AVI 2.1 / CADD 5.4	5'-UTR promoter regulatory variant (c.-1172A>G) in the nuclear vitamin D3 (calcitriol) receptor (OMIM #601769), which regulates intestinal calcium transport and osteoblast gene transcription. **Clinical Impact:** Modulates baseline calcitriol receptor expression; maintain serum 25(OH)D at 35–55 ng/mL via routine cholecalciferol (Vitamin D3) supplementation.
F5	p.Arg534Gln · c.1601G>A · rs6025	Heterozygous	Modifier	AVI 23.6 / CADD 27.9	Heterozygous Factor V Leiden (p.Arg534Gln) impairs Activated Protein C (APC) cleavage of Factor Va, conferring a ~3–5x relative risk of venous thromboembolism (VTE). **Clinical Impact:** Routine lifelong anticoagulation is **not** indicated in asymptomatic heterozygotes; apply situational perioperative/immobilization VTE prophylaxis and avoid systemic prothrombotic estrogen formulations.
PEX6	p.Val788Met · c.2362G>A · rs267608240	Heterozygous	Modifier	AVI 32.3 / CADD 35.0 / REV 0.921 / SpliceAI 0.76	Missense variant (p.Val788Met) in Peroxisomal Biogenesis Factor 6 (AAA ATPase matrix protein importer, OMIM #601498). **Clinical Impact:** Single-allele

Prescribing Guardrails & Annual Preventive Surveillance Schedule

Subject: Daniel Ehrle (Daniel_Ehrle) | **Reference:** GRCh38.p14 (48.8x BAM mean depth)

2. PRESCRIBING & PERIOPERATIVE GUARDRAILS (CPIC LEVEL A & ACUTE RISK TRIAGE)

GENE & VARIANT / DIPLOTYPE	PHENOTYPE / MECHANISM	HIGH-RISK MEDICATIONS / EXPOSURES	PRESCRIBING ACTION & GUIDANCE (CPIC LEVEL A GROUNDED)
F5 Factor V Leiden (rs6025 Het)	APC Resistance	⚠ LIFE-THREATENING RISK **Systemic Estrogens / Oral Contraceptives**	**AVOID / THROMBOEMBOLISM RISK:** Avoid prothrombotic estrogen regimens (35-fold fatal VTE / pulmonary embolism risk amplification); implement situational perioperative VTE prophylaxis (major surgery/immobilization).
CYP2C19 *1/*17 (I331V (rs3758581) A>G, expression (rs12248560) C>T)	Rapid Metabolizer	⚠ LIFE-THREATENING RISK **Voriconazole, Clopidogrel, Mephenytoin, Methadone**	**DOSE_DOWN:** Increased clearance and lower trough concentrations. Perform therapeutic drug monitoring (TDM) to avoid subtherapeutic levels.
CYP2C9 *1/*2 (R144C (rs1799853) C>T)	Intermediate Metabolizer	⚠ LIFE-THREATENING RISK **Celecoxib (Celebrex), Phenytoin (Dilantin), Warfarin (Coumadin), Acenocoumarol**	**DOSE_DOWN:** Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and monitor for GI/cardiovascular adverse effects.
CYP2W1 *1/*6 (P488L (rs3808348) C>T)	Unknown	**CPIC Substrates**	**MONITOR:** Titrate dosing per CPIC.
CYP3A5 *3/*3 (splice defect (rs776746) T>C)	Poor Metabolizer	⚠ LIFE-THREATENING RISK **Carbamazepine, Clopidogrel, Cyclosporine, Efavirenz**	**DOSE_DOWN:** Genotype TT is associated with increased dose of carbamazepine in people with Epilepsy as compared to genotype CC.
DPYD *6/*6 (rs2297595) (c.496A>G / p.Met166Val (rs2297595))	Intermediate Metabolizer (Decreased Pyrimidine Catabolism)	⚠ LIFE-THREATENING RISK **Fluorouracil, Capecitabine, Irinotecan, Oxaliplatin**	**DOSE_DOWN:** Genotypes CT + TT is associated with decreased metabolism of fluorouracil in people with Neoplasms as compared to genotype CC.
NAT2 *5/*5 (I114T (rs1801280) T>C)	Poor Metabolizer	**Isoniazid**	**CAUTION:** NAT2 *5/*5 slow acetylator phenotype. Higher risk of isoniazid-induced hepatotoxicity and peripheral neuropathy. Monitor liver function tests (LFTs) and administer pyridoxine (Vitamin B6).
SLCO1B1 *14/*14 (N130D (rs2306283) A>G, P155T (rs11045819) C>A)	Increased Function	⚠ LIFE-THREATENING RISK **Atorvastatin (Lipitor), Rosuvastatin (Crestor), Simvastatin (Zocor)**	**FAVOUR:** Standard therapeutic efficacy and low myopathy risk.

3. SUGGESTED CLINICAL & LABORATORY SURVEILLANCE SCHEDULE (ANNUAL PHYSICAL)

EVALUATION / LABORATORY TEST	TRIGGERING GENE, VARIANT & ZYGOSITY	SUGGESTED FREQUENCY	CLINICAL RATIONALE & ACTIONABLE THRESHOLDS
Serum Vitamin B12 (Cobalamin), Methylmalonic Acid (MMA), Homocysteine & CBC with MCV Venous Blood Draw	CBLIF c.79+1G>A · rs147785187 chr11:59845374 · DP: 49x Heterozygous (Autosomal recessive)	Baseline at Annual PCP Physical; repeat every 1–2 years	Canonical splice-donor variant (c.79+1G>A) in gastric intrinsic factor (autosomal recessive Hereditary Intrinsic Factor Deficiency, OMIM #261000), which mediates ileal cubam-receptor endocytosis of dietary Vitamin B12. Clinical Impact: Heterozygous carriers preserve intrinsic factor secretion from the wild-type allele; periodic serum Vitamin B12 (> 350 pg/mL) and methylmalonic acid (MMA) verify cellular cobalamin stores (if subclinical depletion occurs, oral/sublingual B12 1,000 mcg/day restores levels via passive mucosal diffusion).
Baseline Pure-Tone Audiometry & Ototoxic	GJB2 p.Met34Thr · c.101T>C · rs35887622	Baseline audiogram at	Coding variant (p.Met34Thr) in cochlear gap junction protein Connexin-26 (DFNB1A, OMIM #220290), which

Clinical Decision Calculus, Evidence Confidence & EHR Integration

4. CLINICAL DECISION CALCULUS: BALANCED INTERVENTIONAL RATIONALE

✅ ARGUMENTS FOR TARGETED PREVENTIVE SURVEILLANCE & PGX

- **Immediate Medication Safety:** Documenting CPIC pharmacogenomic diplotypes prevents avoidable drug toxicity and therapeutic failure before the first prescription is filled.
- **Low-Cost Biomarker Baselines:** Pathway-matched venous blood tests incorporate seamlessly into the annual physical without procedural diagnostic burdens.
- **Proactive Specialist Coordination:** Sharing variant-specific guardrails ensures appropriate situational precautions without over-medicalization.

🛑 ARGUMENTS AGAINST AGGRESSIVE OVER-INTERVENTION

- **Heterozygous Carrier & Modifier Reserve:** Heterozygous variants retain an intact wild-type parental allele (~50–100% capacity) and **do not warrant invasive diagnostic procedures**.
- **Phased Allelic Reassurance:** Whole-genome parental phasing confirms **0 confirmed pathogenic biallelic (trans) recessive compound-heterozygotes**.
- **Polygenic Modifiers ≠ Deterministic Disease:** Modest statistical variants must never trigger unprovoked chronic anticoagulation or off-label polypharmacy.

5. CALCULATED EVIDENCE CONFIDENCE METRICS

EVIDENCE DOMAIN	CALCULATED CONFIDENCE	ANALYTICAL BASIS & CONCORDANCE FACTORS
WGS Analytical Callset	**98%**	48.8x BAM mean depth pan-genome aligned short-read WGS (GRCh38.p14).
Primary Pathogenic Carrier Status (CBLIF, GJB2, VDR)	**95%**	Orthogonal concordance between ClinVar Pathogenic curations and AlphaGenome AVI / CADD scores.
Pharmacogenomic PGx Diplotypes (CYP2C19, CYP2C9, CYP2W1)	**98%**	Dual-engine verified concordance (PharmCAT + Aldy consensus); CPIC Level A clinical guidelines.
Endogenous Protective & Polygenic Risk Alleles	**65%**	Population-level statistical GWAS meta-analyses requiring longitudinal correlation.

6. RECOMMENDED CLINICAL NOTE / EHR PROBLEM-LIST INTEGRATION TEXT

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PATIENT GENOMIC REVIEW SUMMARY (48.8x BAM Mean Depth Phased WGS):
- KEY GENOMIC & CARRIER FINDINGS: CBLIF (c.79+1G>A · rs147785187, Heterozygous); GJB2 (p.Met34Thr · c.101T>C · rs35887622, Heterozygous); VDR (c.-1172A>G · rs4516035, Homozygous); F5 (p.Arg534Gln · c.1601G>A · rs6025, Heterozygous); PEX6 (p.Val788Met · c.2362G>A · rs267608240, Heterozygous). Zero biallelic pathogenic recessive compound-heterozygotes detected; heterozygous carrier/modifier findings retain one wild-type allele and do NOT warrant invasive diagnostic procedures.
- PHARMACOGENOMIC GUARDRAILS (CPIC Level A): Follow standard CPIC dosing. Review patient CPIC diplotypes prior to prescribing; adhere to dosage reduction, drug avoidance, and TDM precautions.
- PREVENTIVE SURVEILLANCE: Incorporate pathway-matched venous blood panels and preventive wellness checks during the routine annual physical.
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